GlucoTwin – Digital Twin for Type-2 Diabetes: system architecture

Team Gym Bro · IIIT Delhi · predicts glucose spikes > 180 mg/dL up to 2 h ahead by fusing EHR + wearable streams

1 DATA SOURCES

Static: EHR (synthetic)

Synthea-style Python generator
FHIR R4 bundles · 200 Indian patients
Demographics, ICD-10 diagnoses
LOINC labs: HbA1c, FPG, insulin, lipids, eGFR, BP · 2-yr history
Medicines (metformin, SU, insulin)
Genetics: TCF7L2, KCNQ1, PRS

Dynamic: wearables + CGM

CGM every 5 min (sensor lag, noise, drop-outs) · steps · heart rate
HRV (RMSSD) · sleep stages
Simulated from a physiological model + Indian daily routines: regional meals, walks, stress, festival sweets, missed doses

Patient app logs

Meals from 35-dish Indian menu (carbs, GI) · medicine doses
Logged imperfectly (as in real life)

Real-world validation

ShanghaiT2DM (Sci Data 2023, CC BY 4.0): 100 people, 15-min CGM, diet + insulin logs, labs

2 INGEST & FEATURES

Ingestion & alignment

FHIR parser -> EHR feature vector
Streams aligned on a 5/15-min grid
Sleep -> nightly summary per day
Steps -> activity intensity
Logged meals -> carb appearance (2-compartment gut model, GI)
Insulin -> s.c. absorption model
Missed metformin -> hepatic output

Feature engineering (t = now)

23 CGM: lags, slopes, variability, time above range, time of day
12 logs: carbs on board, IOB, doses
17 wearables: steps, HR vs rest, HRV, sleep debt, efficiency
24 EHR: HbA1c, HOMA-IR, eGFR, genes
18 twin: simulated 30/60/120 min, peak, insulin sensitivity today

3 DIGITAL TWIN ENGINE

a. Personal physiology

Bergman minimal model + insulin secretion, gut absorption, exercise uptake, sleep -> insulin sensitivity, dawn phenomenon
Prior from EHR (HOMA-IR, HOMA-B)

b. Nightly calibration

Cross-entropy search re-learns insulin sensitivity, meal gain, basal glucose, dawn effect (regularised to EHR prior)

c. Live synchronisation

Observer nudges twin to each CGM
Disturbance state learns unexplained inputs (unlogged snacks, stress)

d. Forecast + what-if

Simulates next 2-6 h from now
Learns routine -> expected meals
What-if: meal, walk, sleep, missed metformin, portion size

twin simulations become features for the ML layer and power the what-if tool

4 AI / ML LAYER

Spike classifier

LightGBM · P(> 180 mg/dL in 2 h)
Threshold tuned on validation patients for 80% sensitivity
Event-level: recall, lead time, false alerts / day

Glucose forecasters

LightGBM 30 / 60 / 120 min
Split-conformal 80% intervals
Clarke error grid evaluation

Explainability

Exact TreeSHAP per prediction -> plain-language reasons
Stream-level attribution

Benchmarks / ablation

Persistence · trend · twin-only
CGM / +EHR / +wearables / fusion / fusion + twin
Deep learning: GRU fusion

5 CLINICIAN DELIVERY

Doctor dashboard

Streamlit + Plotly
Ward view ranked by risk
Patient twin: CGM, forecast band, alert + reasons, wearables, EHR, labs, AGP
What-if simulator
Live replay mode
Model performance

REST API (FastAPI)

/patients · /state
/twin · /what-if
Integrates with EHR (FHIR), CGM / wearable apps

Governance

Synthetic + anonymised data only (DPDP Act 2023)
Patient-level train/val/test
No PII in features
Not a medical device: decision support only